Lecture 32

Sustainable Engineering 5:
Design for Environment

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Applied DfE principles

2 examples from 1st year case study in Introduction to Engineering Materials

Jug Kettle

Electrical tester

Applied DfE principles

2 examples from 1st year case study in Introduction to Engineering Materials

- **Break down into 2 parts:**

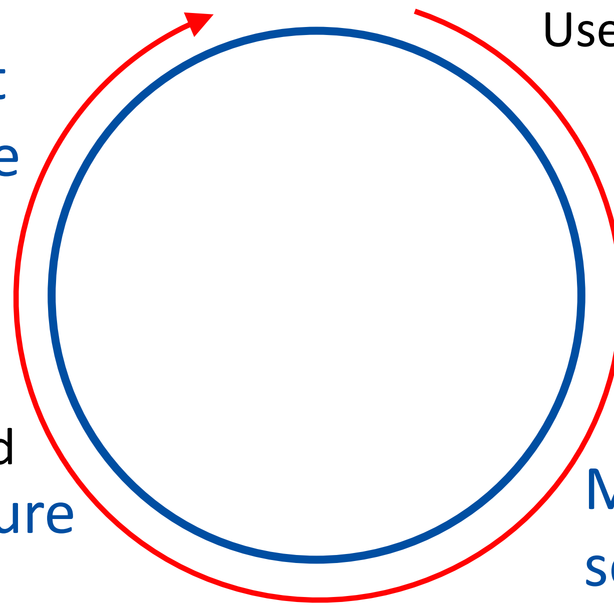
Design cycle

- Product description
- User requirements
- Market position
- Design
    - Materials
    - Manufacture
- Use

Design for Environment

- Materials
- Production
- Transport
- Use
- Recovery

The design cycle



Economics of engineering

Engineering design includes these inter-related aspects:

Marketable

Design

- Meet **customer requirements and business case**
- Materials to achieve performance specifications
- Consider manufacture route

Materials

- Cost of materials
- Supply chain, reliability of source
- Volumes available for manufacture
- Regulation of hazardous materials

Manufacture

- Process steps – cost of production
- Assemble – supply chain of components
- Waste materials

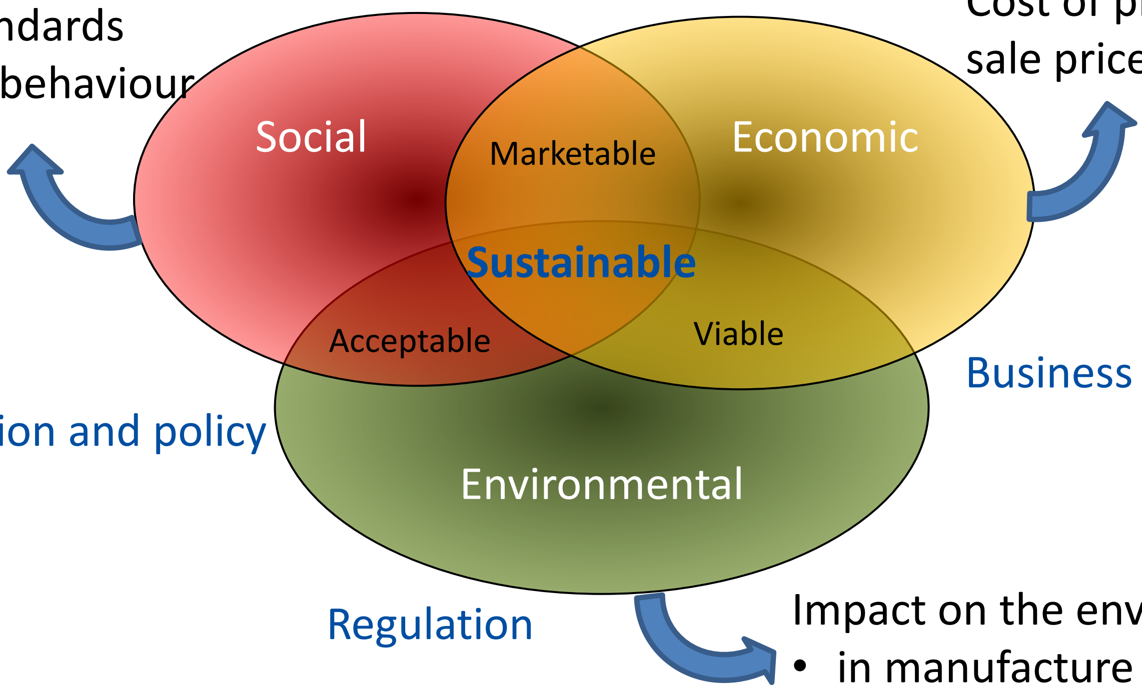
Sustainable engineering

Sustainable engineering requires consideration of 3 factors :

Impact on society
- improvement in living standards
- change in behaviour

Marketplace

Cost of production and sale price must be competitive



Business case

Legislation and policy

Regulation

Impact on the environment
- in manufacture
- in use
- end of live disposal or reuse/recycle

Case 1 – jug kettle

Product description

Specifications:

- 1.5 litre cordless jug kettle
- Power 1800W
- Weight 1.46kg

Price: 99 RMB

User requirements

- Boil water quickly
- Safe to use
- Low energy
- Indicate when hot

Features:

- Auto-shutoff when boiling or over temperature
- Lights up blue when on
- Glass to see contents – capacity indicated
- Easy to clean

Not good features:

- Does not include tea maker
- Low power (most are 3000W)
- Brittle material – could damage

Case 1 – jug kettle

Market position



99RMB product cost is low Affordable for everyone (our product)	159RMB Product cost low Can boil the tea	259RMB The volume is big Not affordable for some people

- Relatively low cost
- Low power
- Good style

Case 1 – jug kettle

Design: components, materials, manufacture

Glass jug

Metal lid and hot plate

Lights

Handle and switch

Electric heater

Separate base, power connection and power cable

Case 1 – jug kettle

Design: components**, materials**, manufacture

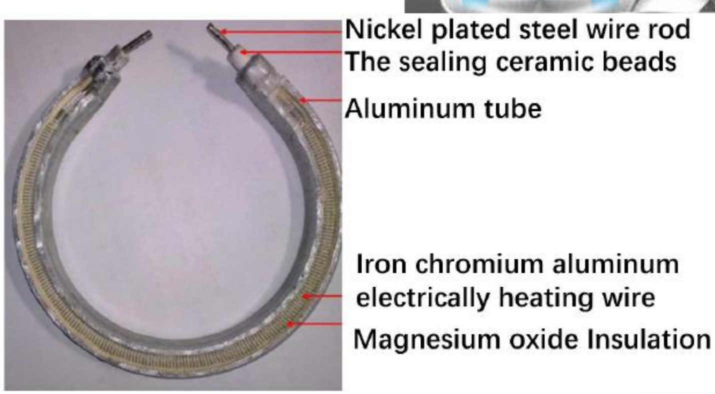
Borosilicate Glass

304 stainless steel

Blue LEDs (GaN)

Polypropylene

Heating element

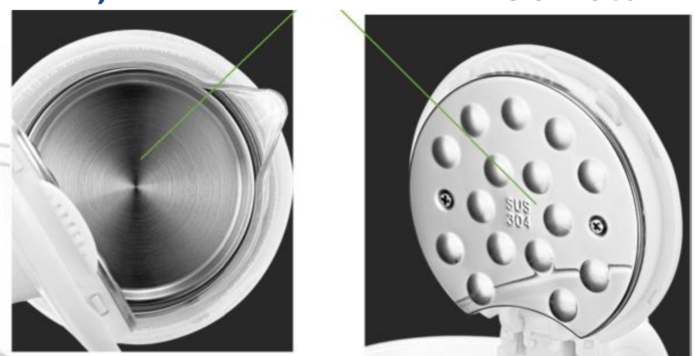


Power connection and power cable Copper – PVC insulation

Case 1 – jug kettle

Design: components, materials, **manufacture**

304 stainless steel
Pressed



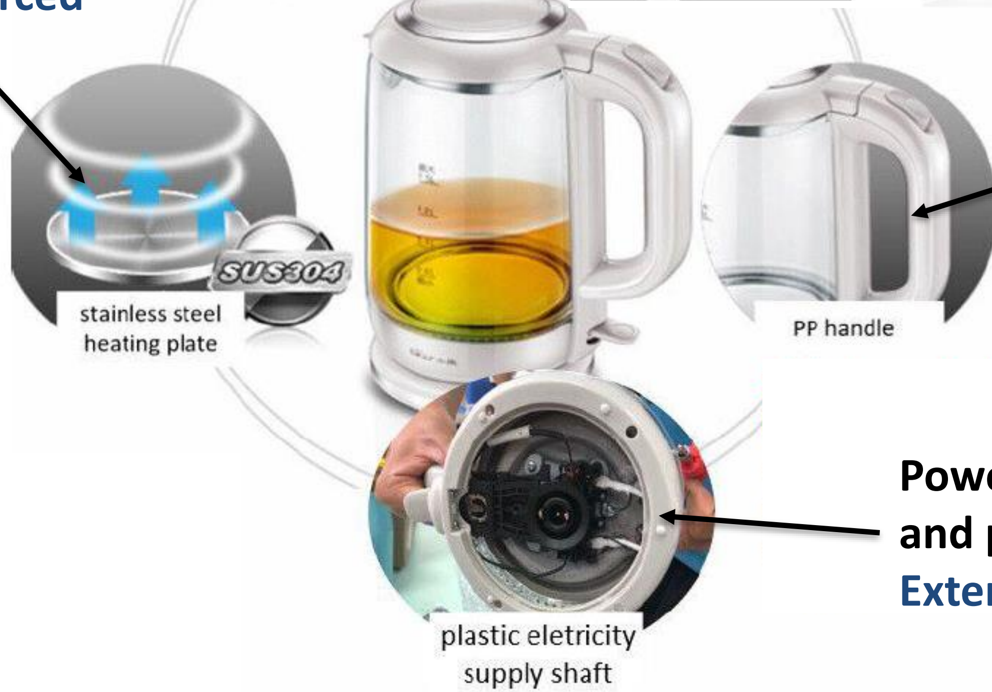
Borosilicate Glass
Blow moulded

Blue LEDs (GaN)
Externally sourced

Heating element
Externally sourced

Polypropylene
Injection moulded

Power connection and power cable
Externally sourced



Case 1 – jug kettle

Assembly:

304 stainless steel
Pressed

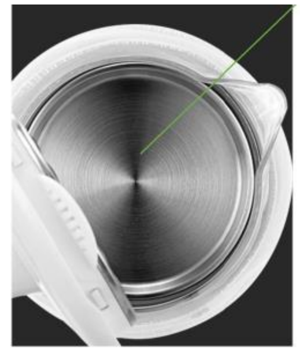
Fixings - screws

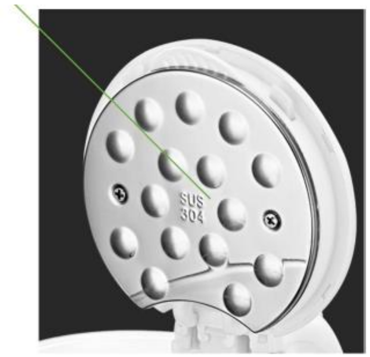
Borosilicate Glass
Blow moulded

Adhesive

Polypropylene
Injection moulded

Snap together





Blue LEDs (GaN)

Heating element

Power connection and power cable
Externally sourced

Case 1 – jug kettle

Use:

Glass container
Stainless steel surface
Are easy to clean
Good chemical resistance
Can see the water boiling
Blue lights when heating

Weight is quite high at 1.46kg

Separate power base is safe to use

Plastic feet can slip
(Elastomer would be better)

Loud noise when boiling

Case 1 – jug kettle

Design for Environment: is it environmentally sustainable?

Materials and their properties

Borosilicate glass
T_m = 1650°C
Softening temperature = 860°C
Thermal conductivity k = 1.4 W/m.K

304 stainless steel
Yield strength σy = 500MPa
Thermal conductivity k = 16.2 W/m.K

Polypropylene
T_m = 130-160°C
T_g = -10°C

Borosilicate glass
Embodied energy = 25 MJ/kg
Carbon footprint = 0.7 kg CO_2/kg
Relative cost 7.9

304 stainless steel
Embodied energy = 80 MJ/kg
Carbon footprint = 0.3 kg CO_2/kg
Relative cost 6.0

Polypropylene
Embodied energy = 100 MJ/kg
Carbon footprint = 3.5 kg CO_2/kg
Relative cost 1.2

Case 1 – jug kettle

Design for Environment: is it environmentally sustainable?

Production and transport

Borosilicate glass
T_m = 1650°C
Softening temperature = 860°C
Thermal conductivity k = 1.4 W/m.K

304 stainless steel
Yield strength σy = 500MPa
Thermal conductivity k = 16.2 W/m.K

Polypropylene
T_m = 130-160°C
T_g = -10°C

Borosilicate glass
Blow mould to final shape
High softening temperature
No finishing needed

304 stainless steel
Only press to simple shape – 2 parts
Fast – high throughput

Polypropylene
Easily injection moulded
6 different parts – high throughput
Low waste
No finishing needed

Case 1 – jug kettle

Design for Environment: is it environmentally sustainable?

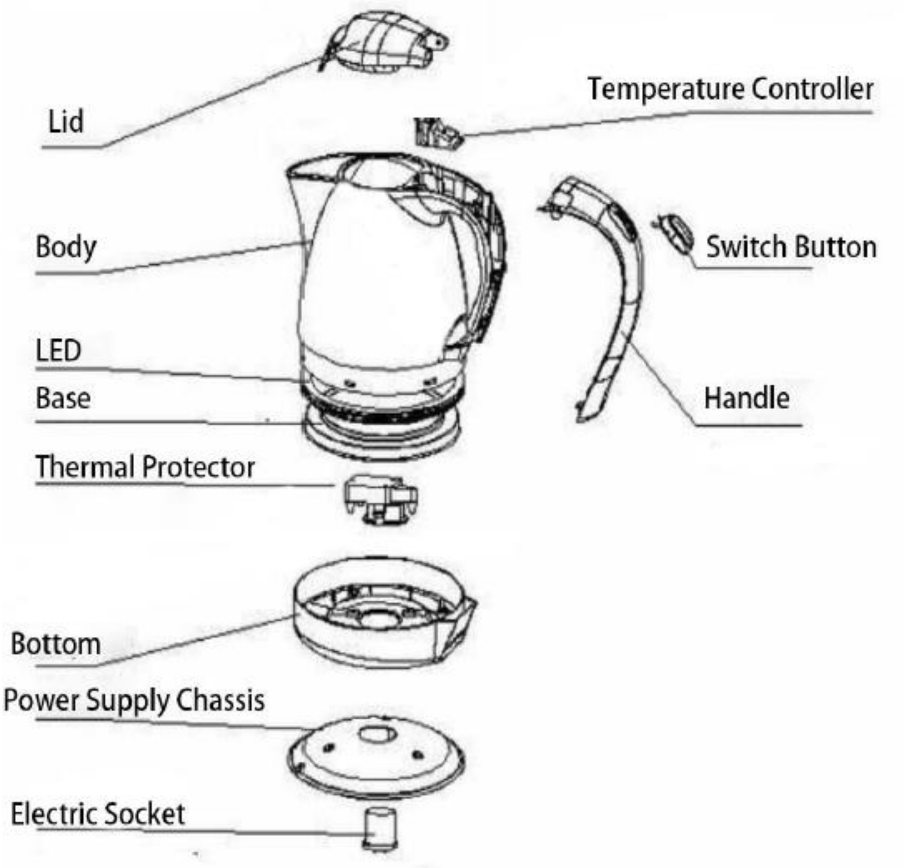
Production and transport

Assembly:

- Not too many parts
- Some out-sourced
- Quick assembly
- Simple tools

Transport:

- Cardboard box
- Total weight ~ 1.5kg



Case 1 – jug kettle

Design for Environment: is it environmentally sustainable?

Use

Power consumption:

- 1800W
- Specific heat capacity of water
  = 4.2 kJ/kg.K
- Energy to boil 1.5 litres
  = 500 kJ

It takes 4.6 minutes to boil
(assuming no heat loss)

Longer time to heat = more heat loss

Cleaning:

- Easy to clean
- Water usage

Lifetime:

- 10 years

Case 1 – jug kettle

Design for Environment: is it environmentally sustainable?

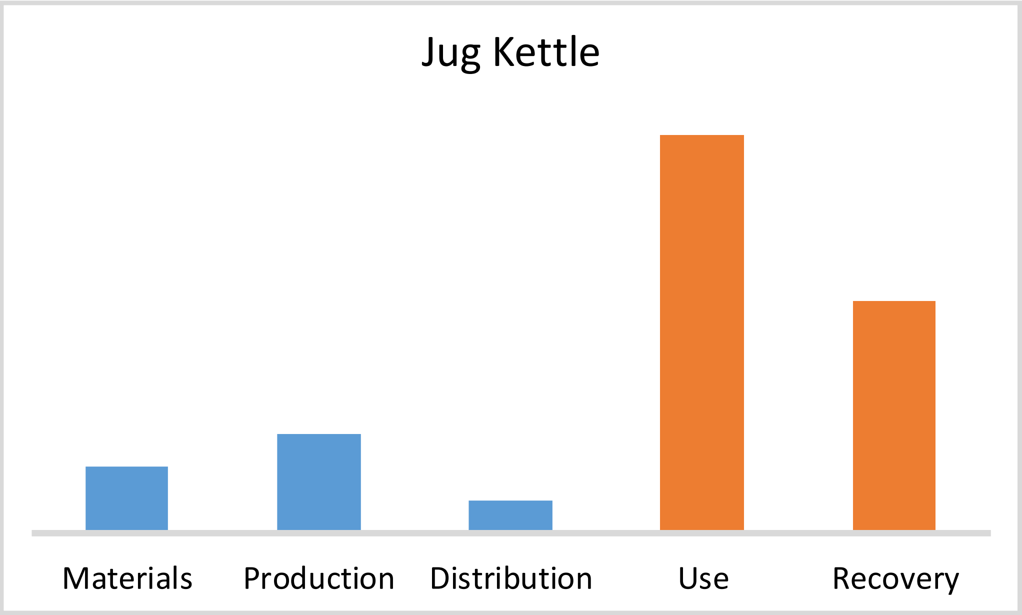
Recovery

- **Materials with low embodied energy**
- **No recycled materials used**
- **No toxic materials used**
- **Difficult to disassemble**
- **Main materials could be recycled**
  - But difficult to separate
- **Production energy low**
- **Little waste material**
- **Recycleable packaging**

Case 1 – jug kettle

Design for Environment: is it environmentally sustainable?

Product impact

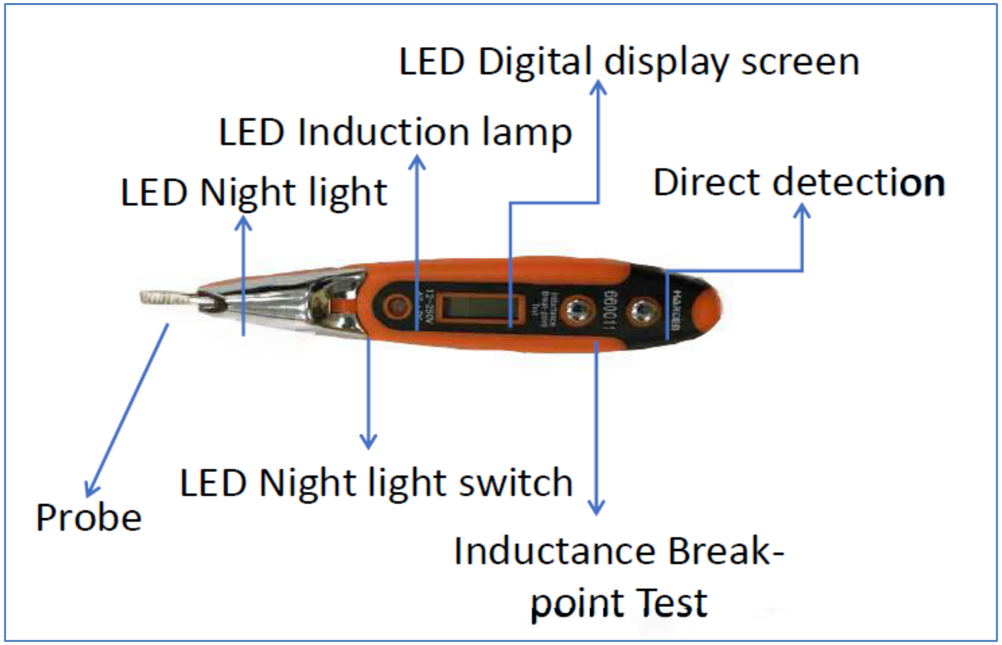


Product design and manufacture are relatively low impact

Use and disposal of product are poor

Case 2 – electrical tester

Product description



Specifications:

- Electrical circuit tester
- Multifunctional
- Non-professional use

Price: 23 RMB

User requirements

- Domestic (home) use
- Test electrical circuits
- Convenient
- Long lasting

Features:

- Tough construction
- Pocket sized
- Multifunctional
- Light for working in difficult places

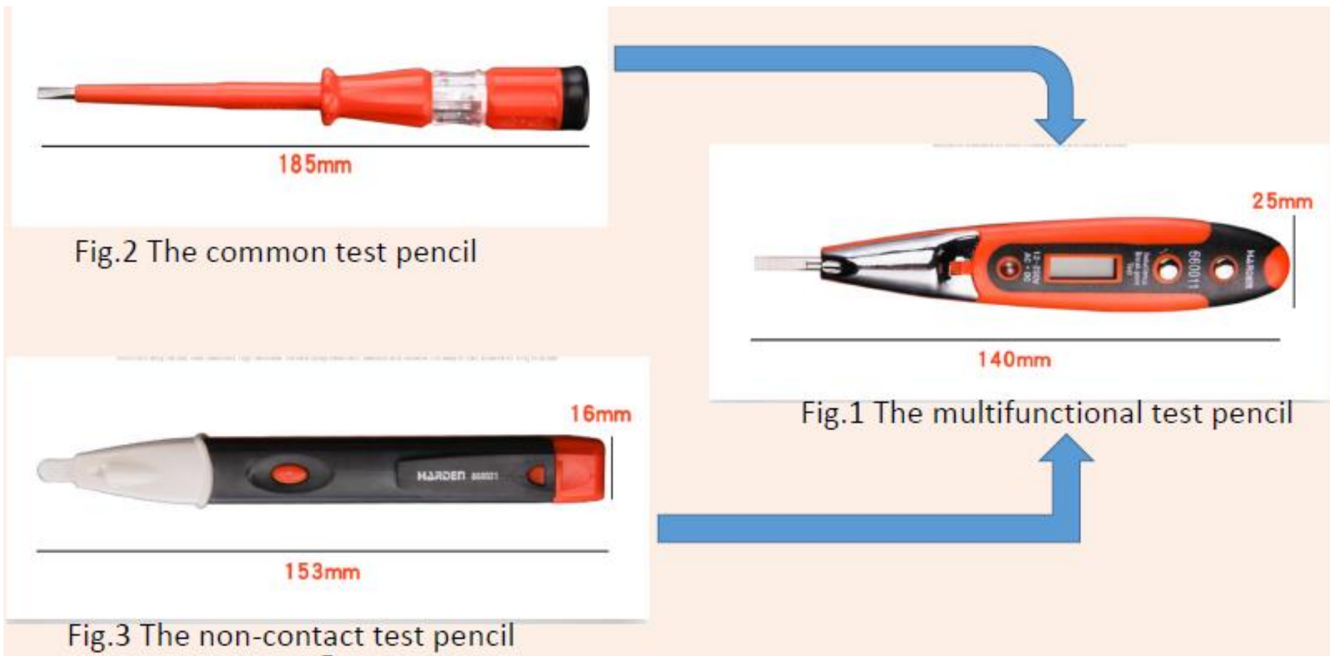
Not good features:

- Limited range of voltage measurement
- Battery consumable (not recharged)

Case 2 – electrical tester

Market position

7.8 RMB



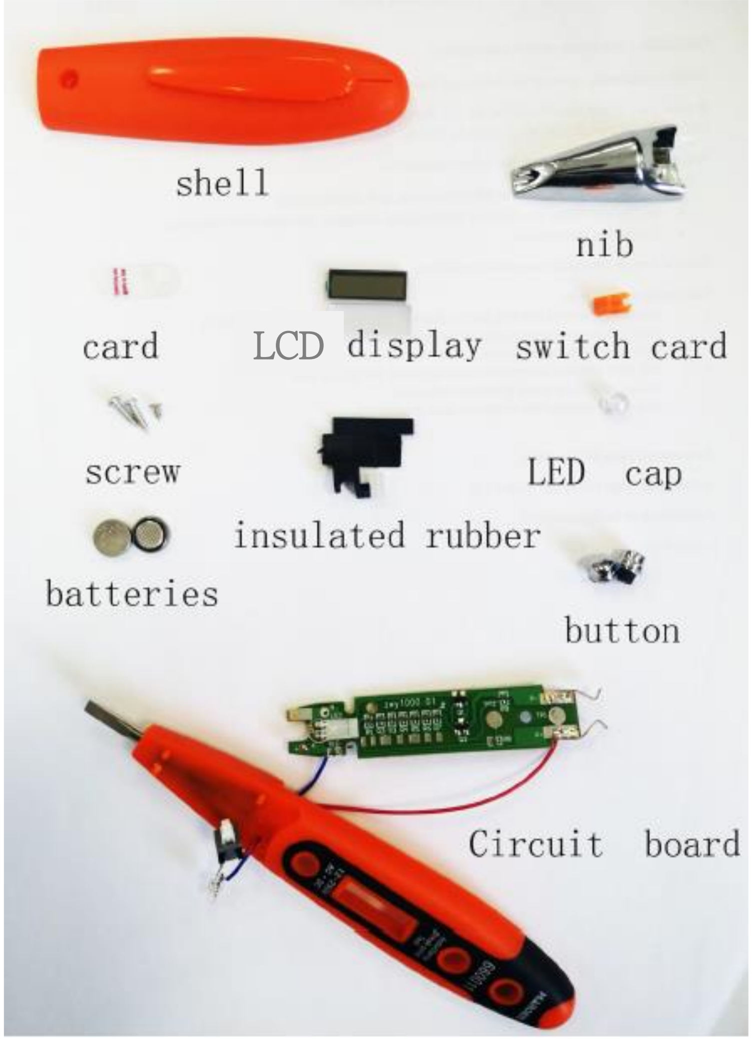
23 RMB

43 RMB

- Mid price
- Small size
- Good functionality

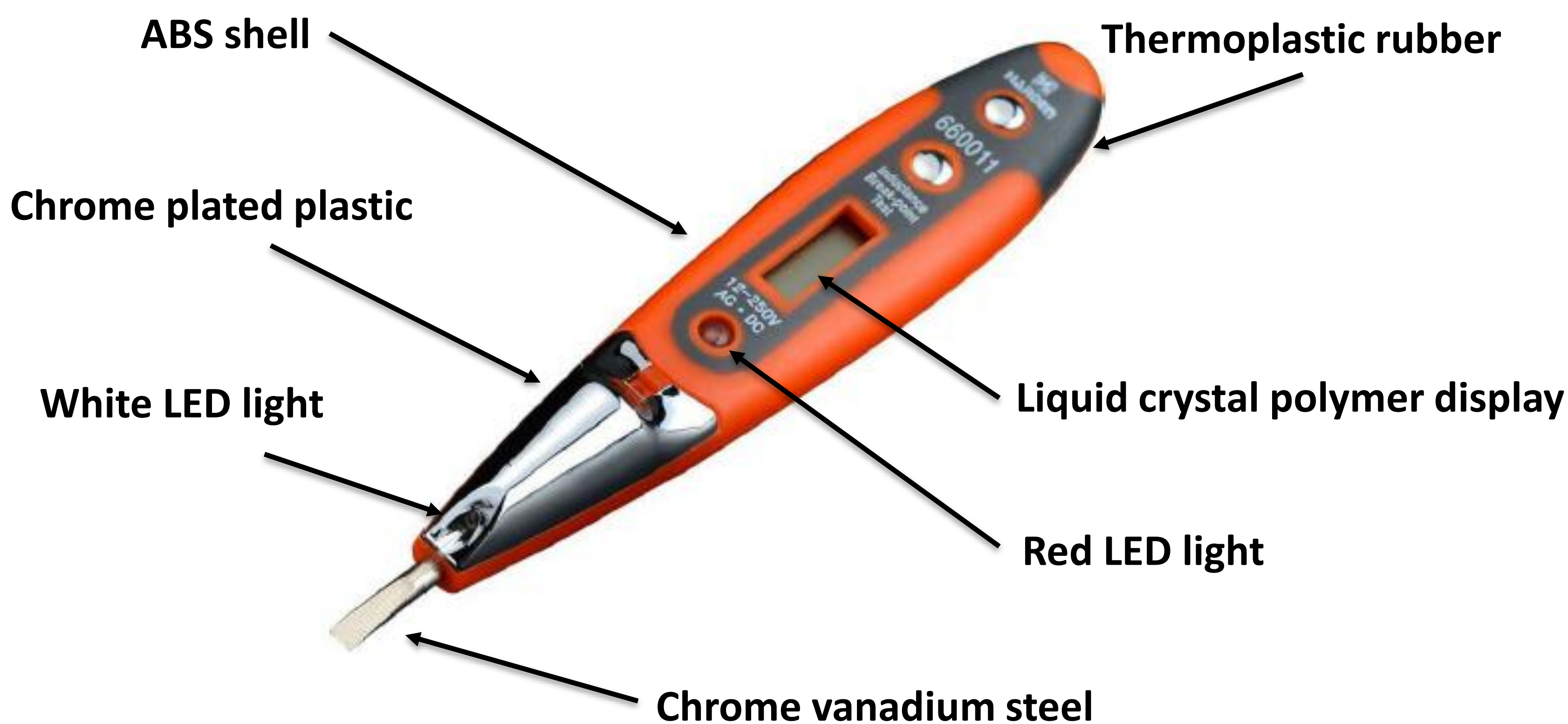
Case 2 – electrical tester

Design: components, materials, manufacture



Case 2 – electrical tester

Design: components**, materials**, manufacture



Case 2 – electrical tester

Design: components**,** materials, **manufacture**

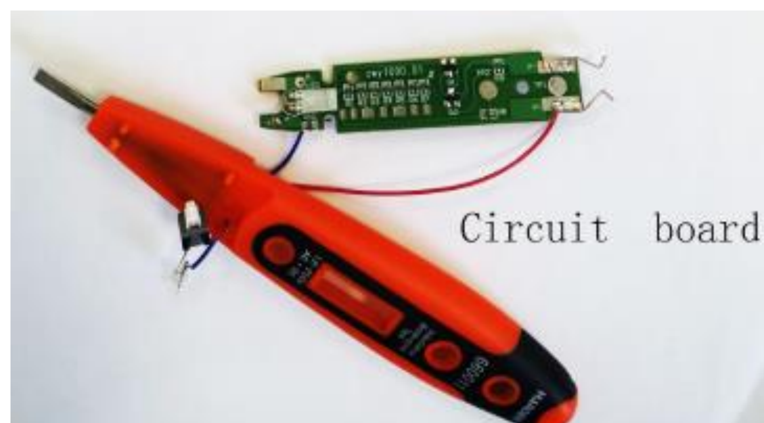
Metal probe
Forged

Thermoplastic rubber
Injection moulded

ABS shell
Injection moulded

LED light
Display panel
Circuit board
Battery
Switch

Externally sourced



Case 2 – electrical tester

Assembly:

Wires
Soldered in place

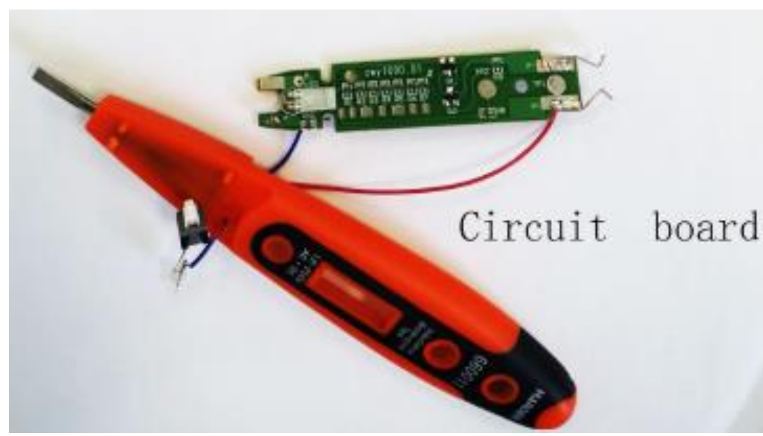
Electrical components
Fit in place

ABS shell
Injection moulded

1 Fixing screws

LED light
Display panel
Circuit board
Battery
Switch

Externally sourced



Case 2 – electrical tester

Use:

ABS shell
Cr-V steel probe
Tough
Hardwearing

Size is quite small at 140mm
Easy to fit in pocket
Simple to switch functions

Uses Silver oxide battery
Long lasting but expensive to replace

Lights up for work in dark places

Limited voltage measurement range

Case 2 – electrical tester

Design for Environment: is it environmentally sustainable?

Materials and their properties

ABS
T_m = 200°C
T_g = 105°C

Thermoplastic elastomer
T_m = 190°C
T_g = -40°C

Cr-V steel
Yield strength σy = 900MPa
Electrical resistivity 8×10^{-7} Ω.m
Good corrosion resistance

ABS
Embodied energy = 111 MJ/kg
Carbon footprint = 4 kg CO_2/kg
Relative cost 1.2

Thermoplastic elastomer
Embodied energy = 74 MJ/kg
Carbon footprint = 0.3 kg CO_2/kg
Relative cost 4.0

Cr-V steel
Embodied energy = 30 MJ/kg
Carbon footprint = 1.9 kg CO_2/kg
Relative cost 1.6

Case 2 – electrical tester

Design for Environment: is it environmentally sustainable?

Materials and their properties

ABS
T_m = 200°C
T_g = 105°C

Thermoplastic elastomer
T_m = 190°C
T_g = -40°C

Cr-V steel
Yield strength σy = 900MPa
Electrical resistivity $8 \times 10^{-7}\ \Omega.m$
Good corrosion resistance

ABS
Easily injection moulded
3 different parts – high throughput
Low waste
Electroplated chrome on 1 part

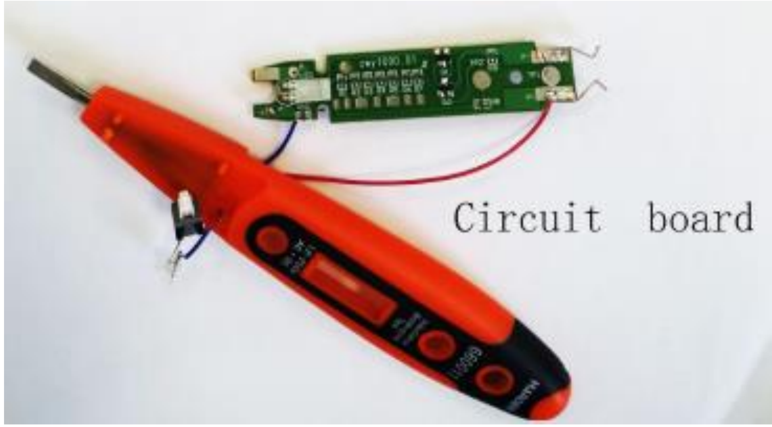
Thermoplastic elastomer
Injection moulded – high throughput
Low waste
No finishing needed

Cr-V steel
Press to simple shape – high pressure
Fast – high throughput

Case 2 – electrical tester

Design for Environment: is it environmentally sustainable?

Production and transport



Assembly:
- 12 parts
- Some out-sourced
- Manual assembly
- Solder wires

Transport:
- Card and blow moulded cover
- Packaging recyclable
- Very light weight

Case 2 – electrical tester

Design for Environment: is it environmentally sustainable?

Use

Power consumption:
- Silver oxide batteries
- Not rechargeable
- Liquid crystal display
- LED lights

Low power consumption

Materials used:
- Tough
- Durable

Lifetime:
- 30 years

Case 2 – electrical tester

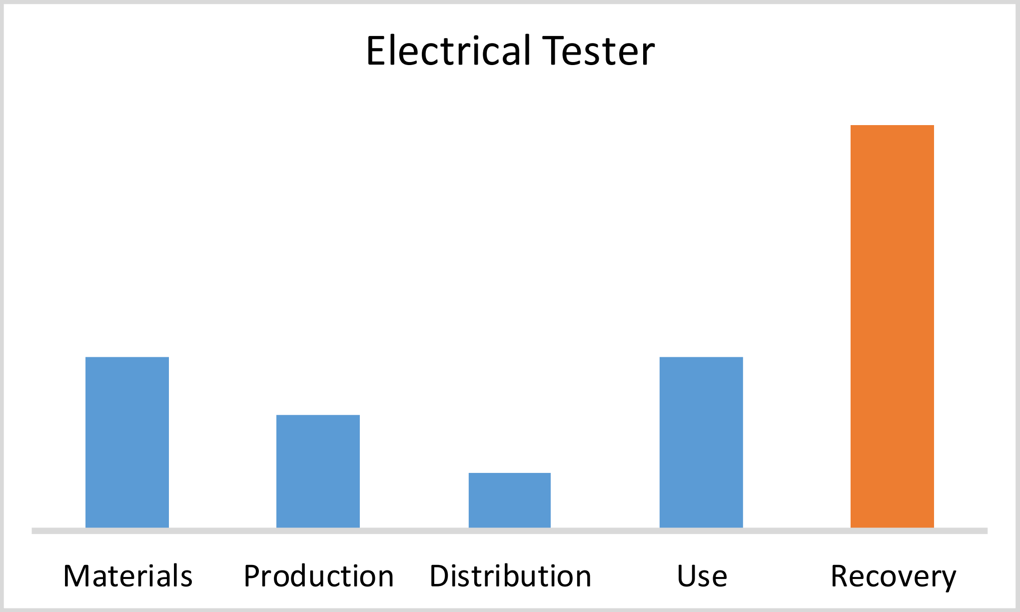
Design for Environment: is it environmentally sustainable?

Recovery

- **Materials with low embodied energy**
- **No recycled materials used**
- **Some toxic materials used (Chrome)**
- **Not difficult to disassemble**
- **Main materials could be recycled**
  - But e-waste from electronics
- **Production energy low**
- **Little waste material**
- **Recycleable packaging**

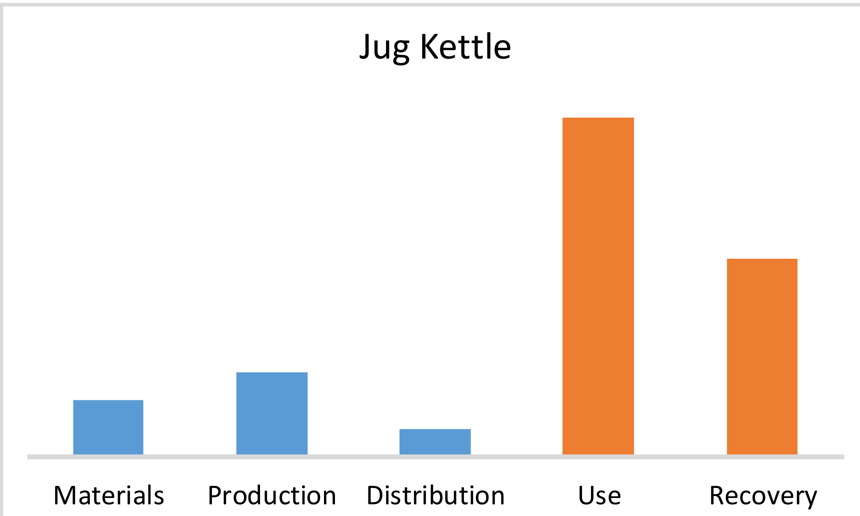
Case 2 – electrical tester

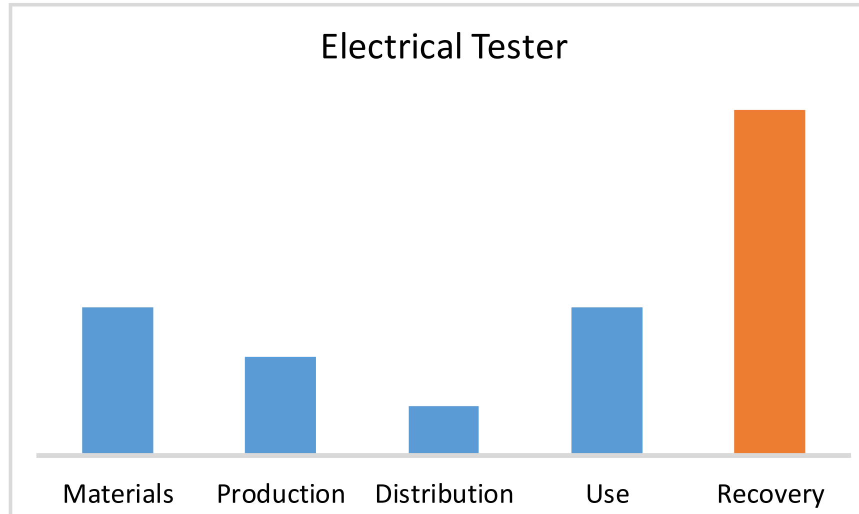
Design for Environment: is it environmentally sustainable?
Product impact



Product design and manufacture are relatively low impact
Disposal of product is poor = e-waste

DfE assessment?





Can we say that these products are sustainably engineered?

The design and manufacture is relatively low impact

But the disposal planning is environmentally un-friendly

DfE assessment?

Can be defined as:

“Sustainable development is : development that meets the needs of the present without compromising the ability of future generations to meet their own needs”

Report of the Brundtland commission of the UN, 1987